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# Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation

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ABSTRACT

Geometry-conditioned multi-view diffusion supplies images for 3D texturing, but useful control strength depends on the implementation and evaluation endpoint. We study Timestep-Conditioned Adapter Scaling (TCAS) on an existing GeoTex-Adapter residual path, retaining C3 as a historical middle-high candidate. A complete 300-object equal-applied-scale-budget comparison finds that C3 trails fixed mean by 0.13 dB and late-high by 0.56 dB FG-PSNR; fixed-high wins this endpoint on 172 objects and C3 on none. These controlled allocation effects do not establish a residual-dose-independent mechanism. On ten previously unbaked, compatibility-selected objects, calibrated native-UV baking with eleven unseen views per object yields a Late-high–Fixed Mean FG-LPIPS difference of −0.00265 (95% object-bootstrap CI [−0.00414, −0.00137]), favorable on all ten. C3's PSNR advantage over Fixed High does not repeat stably and its SSIM is lower. Generated-view and unseen-bake rankings are broadly positively associated, although the LPIPS winner changes on seven objects. A separate fifty-object native MV-Adapter comparison gives Middle-high–Fixed Mean FG-PSNR +0.029 dB ([−0.019, +0.081]), without a separated primary gain. Scheduling is established prior work; we withdraw independent CAI derivation and optimality claims. The contribution is controlled, task-specific evidence, a bounded allocation result and explicit cross-interface limits, not a universally optimal schedule or validated PBR fidelity.

## 1. Introduction

Multi-view diffusion systems generate target-view images that can later be projected or baked onto a 3D surface. In geometry-conditioned systems, an adapter injects shape information into the denoiser through intermediate residuals. The strength applied to those residuals is therefore a practical inference-time control: changing it can alter how strongly geometry conditions the generated views and can also change their appearance. The useful setting depends on the adapter implementation and on the image endpoint being considered, so a single structural score cannot characterize the full effect.

This paper studies that control question in a fixed MVPainter-style multi-view texturing pipeline. Timestep-Conditioned Adapter Scaling (TCAS) applies a piecewise requested-scale profile to the existing GeoTex-Adapter residual path without training parameters or changing model weights. The formulation records three separate quantities: the scale requested for a denoising step and layer group, the scale actually applied after native wrapper caps, and the norm of the scaled residual correction. This makes the implementation and the scope of each comparison measurable.

Layer- and timestep-dependent control is established prior work. Scheduled Style Injection (SSI) varies StyleID's attention-mixing parameter across decoder layers and denoising steps, and also schedules geometry-conditioned ControlNet strength along those axes. We therefore do not present TCAS as a new general scheduling primitive. Our contribution is an empirical graphics-system characterization of a specific residual path: we measure how requests pass through native caps, how the applied corrections vary, and how fixed and scheduled controls compare on paired multi-view image endpoints. SSI's style-transfer results do not determine the results of these object-level texturing comparisons; differences in task and interface require measurement here, but do not make the underlying scheduling idea new.

We compare the recorded C3 profile with stronger fixed and

scheduled controls on 300 held-out Fresh C objects. Equal-applied-budget stage allocation favors fixed mean and late-high over the historical middle window. We then test a named late-high candidate on ten previously unbaked objects under calibrated native-UV projection, and the middle-window hypothesis on fifty previously uninspected MV-Adapter objects using native-camera targets. The new bake pilot supports a small Late-high–Fixed Mean LPIPS benefit, but not a stable C3–Fixed High PSNR benefit. The second-interface primary does not separate. Thus useful strength allocation is conditional on interface and endpoint, rather than a universal C3 rule.

The contributions are:

(1) **Controlled stage-allocation evidence.** A complete six-condition, 300-object equal-applied-budget comparison measures the total effect of stage placement and supplies strong fixed controls (Table 6). It rejects a generally preferred middle window; activation-dependent residuals limit further mechanism attribution.

(2) **A bounded texturing deployment result.** A calibrated ten-object, seven-condition native-UV bake pilot evaluates eleven unseen views per object and supports the named Late-high–Fixed Mean FG-LPIPS contrast, while retaining C3 counterevidence and actual appearance failures (Table 9, Figure 8). Positive rank association and changes of the winning condition are reported separately.

(3) **Native-interface validation and reproducibility.** A complete fifty-object MV-Adapter comparison with native-camera targets and default-path identity checks tests the middle-window hypothesis on a second interface (Table 11). Requested/applied/residual accounting, paired records, calibration tests and delivered analysis code support these empirical conclusions. We add no learned branch, selector or general scheduling primitive; historical preference, FAC and cross-interface records retain their original limits.

## 2. Related Work

### 2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. Single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have improved structural stability and view synthesis [1, 2, 3, 4, 5, 6, 7, 8]. Texture synthesis remains a distinct challenge for asset realism and usability [15]. Many systems generate RGB views with 2D diffusion and project or bake them onto a surface [9, 10, 11, 12]. Recent work also explores exemplar-based texture synthesis and diffusion-based procedural materials [16, 17].

Paint3D, MVPaint, Hunyuan3D 2.0, and MVPainter use geometry-conditioned multi-view generation in 3D texturing pipelines [18, 19, 20, 13]. MV-Adapter, MVDiffusion, and other synchronized multi-view methods address consistency through different adapter and denoising designs [21, 22, 23, 24]. These systems establish the graphics context for this work. The present evidence does not compare final baked-texture quality across those systems.

### 2.2. Adapter-Augmented Diffusion Models and Geometric Control

ControlNet and T2I-Adapter provide learned spatial-conditioning branches or feature injections, whose contribution can be weighted by an inference-time condition-strength scale [25, 26]. They establish conditioning strength as a practical control, but their injected tensors and scaling sites do not share a numerical dose with GeoTex. The GeoTex-Adapter path in the 01549 study already exposes an inference-time residual scale and evaluates fixed requests. TCAS reuses that control point as a three-stage request over sampling progress, while recording how wrapper-group caps change the applied values and how adapter activations determine the resulting corrections. It adds no learned branch or schedule selector. IP-Adapter introduces image prompts through decoupled attention [27]; LoRA and related methods use scaled low-rank updates [28, 29, 30].

### 2.3. Diffusion Guidance Scheduling and Timestep Control

Diffusion denoising is often discussed in terms of coarse-to-fine evolution, and prior work varies guidance strength over sampling progress to study quality, diversity, or fidelity [32, 33, 34, 35]. Limited-interval classifier-free guidance is one example; it changes the combination of conditional and unconditional score predictions [14].

SSI is the closest scheduling precedent. It varies StyleID's query-mixing parameter across decoder layers and denoising time, tests schedule direction and shape, and applies scheduling to a separate geometry-conditioned ControlNet branch; its reported ControlNet scheduling gains over fixed conditioning are modest relative to its style-injection results [31]. This prior work already covers the general idea of layer/time scheduling and geometry-conditioning schedules. TCAS acts on the residual correction produced by GeoTex-Adapter from normal, depth, and foreground-mask conditions, with group-specific wrapper caps; it does not modulate StyleID attention mixing or add a ControlNet branch. The task, interface, and measured endpoints differ, so SSI does not imply the Fresh C paired estimates, the requested/applied/residual accounting, or the observed image- and color-agreement failures. These task-specific measurements are the empirical contribution; the scheduling concept itself is not claimed as new.

### 2.4. Multi-view Adapter Interfaces

MV-Adapter uses a condition guider for camera or geometry maps together with decoupled spatial, multi-view, and optional image cross-attention layers [21]. MVDiffusion uses correspondence-aware attention modules to connect pixels across views [22]. These interfaces differ from the GeoTex-Adapter residual branch studied here. We therefore report each auxiliary intervention within its own tested implementation and cohort, without pooling metrics or interpreting it as an architecture-matched replication of TCAS. The main cross-interface results are summarized in Table 10.

## 3. Method

### 3.1. Task Definition and Base Pipeline

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [13] and studies how the inference-time strength of geometric condition injection affects the generated results. Figure 1 shows the TCAS control gate on the GeoTex-Adapter residual path: the requested scale is limited by the wrapper's group-specific cap before the residual is added to frozen UNet features. Given a single reference image and the corresponding 3D geometry, the base pipeline generates RGB texture images from six target views and can pass them to downstream projection or texture baking. Fresh C evaluates target-view images; the separate ten-object pilot evaluates calibrated unlit baked meshes. We do not modify the base UNet, VAE, or pretrained adapter weights.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$\mathbf{G} = \{G_{\text{normal}}, G_{\text{depth}}, G_{\text{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For inference step $j$ with normalized execution progress $p_j$, the adapter injection can be written as:

$$\mathbf{h}'_{j,\ell} = \mathbf{h}_{j,\ell} + s_{\text{app}}(p_j, \ell)\, A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G}), \tag{2}$$

Here $\ell$ indexes an injected layer group. The *requested scale* $s_{\text{req}}(p_j, \ell)$ is the value assigned by the profile; the *applied scale* is the value used by the wrapper,

$$s_{\text{app}}(p_j, \ell) = \min(s_{\text{req}}(p_j, \ell), c_\ell), \tag{3}$$

when the native cap $c_\ell$ is active, and otherwise equals the request. The resulting correction is $\mathbf{r}_{j,\ell} = s_{\text{app}}(p_j, \ell) A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G})$; its *post-scale residual magnitude* is $m_{j,\ell} = \|\mathbf{r}_{j,\ell}\|_2$. Thus neither a requested nor an applied scale label alone determines the correction magnitude.

In the audited Fresh C wrapper, the three cap groups map the GeoTex residual wrappers on the UNet ResNet blocks as follows: deep comprises the UNet middle block and `up_0`, middle is `up_1`, and shallow is `up_2`. These names identify the implementation's grouped injection points; they do not assert universal semantic roles for denoising stages. The corresponding Fresh C caps are 3.0, 3.5, and 0.8. C3's shallow request of 1.25 is therefore applied as 0.8, while its deep and middle requests pass unchanged. GFL and GFH also clip at the shallow cap. These cap values describe the audited Fresh C wrapper; row-level cap semantics for the historical 24-object probe are not fully recoverable. TCAS scales a geometry residual added to feature activations; it does not reweight conditional and unconditional score predictions as classifier-free guidance does.

### 3.2. Image-Space Metrics and Texture-Variation Diagnostics

We use separate terms for separate outcomes. *Image-space agreement* refers to target-view PSNR, LPIPS, and SSIM measurements on the named image regions; it is not a measure of final 3D material quality. *Reference-color agreement* refers to color differences such as CIEDE2000 computed against the corresponding reference. *Texture variation* refers to statistics such as RGB standard deviation, Laplacian variance, gradient magnitude, and high-frequency energy; these may rise with detail or artifacts and are not fidelity measures. *Subjective preference* refers only to the submitted forced-choice study. *Baked 3D appearance* refers to rendered mesh outputs after texture projection or baking, while *PBR/material fidelity* additionally concerns appearance under the material and illumination model. These endpoints are not interchangeable. FG-SSIM, Edge-SSIM, and Crop-SSIM quantify structural similarity on the indicated image regions; PSNR measures pixel error; LPIPS compares learned image features [36, 37].

Laplacian variance, RGB standard deviation, and gradient magnitude can describe local high-frequency or color variation. For example, a foreground Laplacian statistic may be written as

$$M_{\text{var}} = \text{Var}(\text{Lap}(I_{\text{out}} \odot G_{\text{mask}})), \tag{4}$$

where $G_{\text{mask}}$ is the foreground mask. A larger value is not necessarily better: detail, noise, repeated patterns, or artifacts can all increase local variation. We therefore do not interpret these proxies as texture fidelity or use them alone to claim that a schedule better preserves reference appearance.

### 3.3. TCAS Control Semantics and Comparison Design

TCAS replaces a globally fixed requested adapter scale with a piecewise profile over normalized sampling progress. The early/middle/late partition is an operational description, not a universal claim about the function of every denoising stage. C3's low–high–low profile is the specific development choice evaluated in this revision; it is not presented as an independently derived optimum.

To avoid ambiguity caused by different raw timestep directions in different schedulers, $p_j = j/(T-1)$ denotes execution progress for inference step $j \in \{0, \ldots, T-1\}$. Thus $p = 0$ is the first update from the initial noisy latent and $p = 1$ is the last denoising update before decoding. For $T = 50$, TCAS divides execution into steps 0–16, 17–32, and 33–49 (17/16/17 steps):

$$s_{\text{req}}(p) = \begin{cases} s_e, & 0 \le p < 1/3 \\ s_m, & 1/3 \le p < 2/3 \\ s_l, & 2/3 \le p \le 1 \end{cases}. \tag{5}$$

We call this variation across the 50 inference steps the *temporal allocation*. A *layer allocation* varies the requested scale across the three wrapper groups at a fixed step. For a schedule with applied values $s_{\text{app}}(j, \ell)$, its per-group applied-scale sum is $B_\ell = \sum_j s_{\text{app}}(j, \ell)$; the unweighted sum $B_{\text{nom}} = \sum_\ell B_\ell$ is a *nominal applied-scale budget*. It does not weight scales by adapter activations. We log post-scale correction summaries, including wrapper-level $L_2$ norm and mean absolute entry. The three wrapper groups operate on feature maps with different spatial dimensions, and the traces do not retain channel counts; raw $L_2$ values are therefore not used for cross-layer dose comparisons. The per-entry mean absolute correction reported in Supplementary Section S4 is a dimension-normalized descriptive

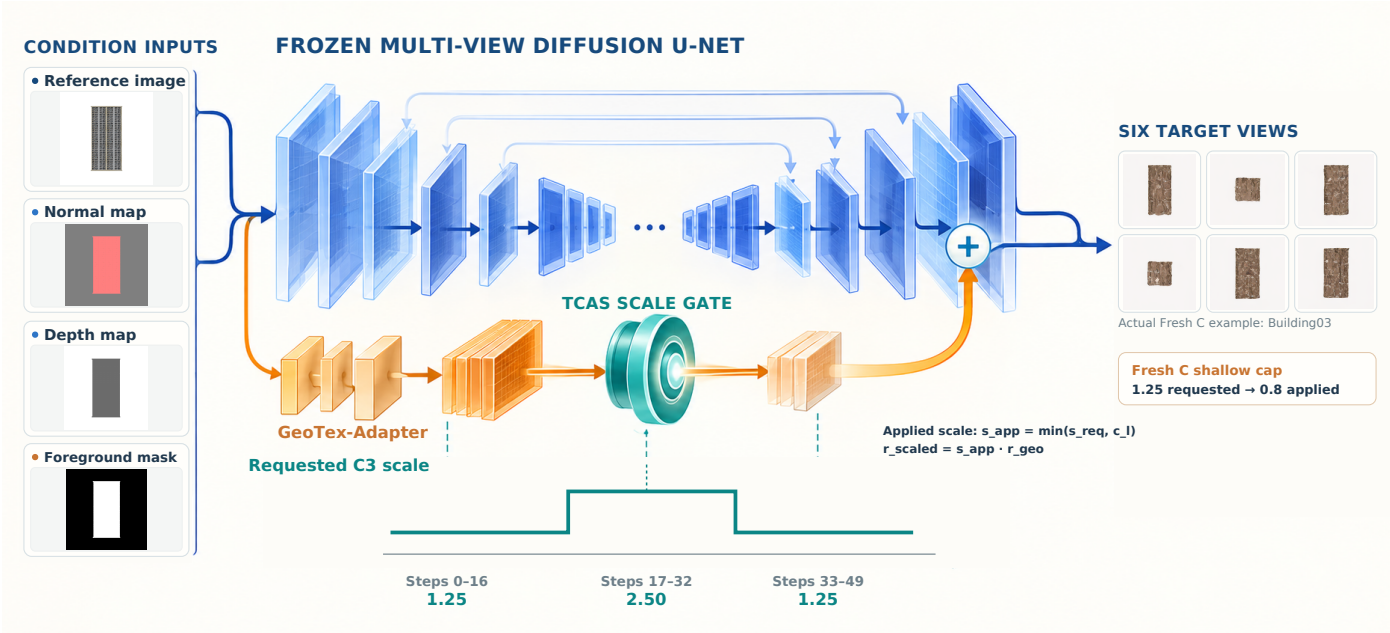



**Fig. 1. TCAS on the GeoTex-Adapter residual path. The requested C3 profile is** $(1.25, 2.50, 1.25)$ **over steps 0–16, 17–32, and 33–49; the wrapper applies the request after group-specific native caps, then adds the scaled residual to frozen UNet features. In Fresh C, the shallow-group cap is 0.8. Input maps and six C3 output thumbnails are actual Fresh C Building03 data (Shalmon, CC BY 4.0; Supplementary Table S1), used here to illustrate the interface. Conceptual artwork was generated with OpenAI Images (version not recorded by the tool) and assembled with reproducible vector labels; it does not encode measured results. Baking is evaluated separately in Section 4.9.**

summary, not an equal-dose intervention. Correction magnitudes also depend on the current hidden state and earlier denoising updates, so equal nominal budgets do not imply equal realized corrections.

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{6}$$

The low–high–low profile is a concrete inference-time intervention motivated by the submitted development analysis. Its stage interpretation is a task-specific hypothesis: the tested early, middle, and late allocations may produce different responses. Fresh C evaluates the recorded profile against matched controls but does not prove a universal function for any stage. TCAS adds no trained parameters and does not modify base-model or adapter weights; it replaces the requested scale during inference.

*What C3 changes and what its comparisons identify.* C3 requests $(1.25, 2.50, 1.25)$ for each of the 50 steps across the three layer groups before native caps. In Fresh C, the applied deep and middle profiles retain those values; the shallow profile is clipped to 0.8 throughout. GFL requests 1.25 at every step and group, so its deep and middle values are 1.25 throughout and its shallow value is also clipped to 0.8. Their per-group applied-scale sums are therefore 82.5 versus 62.5 in deep and middle, and 40 for both in shallow. The C3–GFL contrast changes the middle-third deep/middle scale magnitude as well as its temporal allocation; it is not a timing-only comparison.

To identify a temporal-placement contrast, controls must hold the per-layer applied values and nominal per-group sums fixed while moving the same values between time windows. To identify a scale-strength contrast, controls must hold the layer and temporal allocations fixed while changing amplitude. Both contrasts also require reporting post-scale correction summaries: matching requested values or applied-scale sums does not match the evolving residual corrections. Raw $L_2$ values are dimension-sensitive across wrapper groups and cannot establish cross-layer dose matching. The current 300-object allocation comparison exactly matches the applied sums in each layer group. This controls the scalar budget for the total allocation effect, although it does not identify a residual-dose-independent mechanism. Post-hoc adjustment for state-dependent corrections cannot by itself recover that mechanism. The older strict-276 analysis is reported separately.

*Development origin and scope of C3.* The submitted 01549 paper records C3 = $(1.25, 2.50, 1.25)$ as the selected profile from a 24-object development probe containing 17 uniform, layer-wise, and time-conditioned variants. We retain that development record because it explains the origin of the profile studied here. The complete row-level outputs and selection-time provenance are not available in the current evidence authority, so the historical ranking and labels cannot be independently checked. We therefore make no formal optimality or uniqueness claim based on this probe. Fresh C evaluates the recorded profile without further schedule search.

### 3.4. Geometry Adapter and Inference Interface

GeoTex-Adapter is treated as the fixed geometry-residual backbone, not as a new contribution of this paper. It receives a five-channel input comprising a three-channel normal map, a one-channel depth map, and a one-channel foreground mask. A

multi-scale encoder produces residual features that are injected into the frozen UNet. TCAS acts on the requested multiplier along denoising progress; it does not retrain the adapter, the UNet, or the VAE and introduces no learned parameters.

The first-round 01549 revision also reported a learned Full Adaptive Correction (FAC) extension. We retain its controlled negative re-examination as a previously reported auxiliary result in Section 4.12 and Supplementary Table S16; it is not a new 1008 training run. TCAS remains the paper's sole training-free method.

## 4. Experiments

### *4.1. Experimental Setup*

#### *4.1.1. Experimental protocol and comparisons*

The original 01549 submission reports fixed-scale experiments, a 24-object schedule-development probe, and a 300-object pooled evaluation. The source figures and tables are authentic submission artifacts, but the current evidence archive does not contain a complete object-level manifest and run record for every historical result. We therefore preserve the fixed-scale and C3-selection questions as historical development evidence and do not present their unrecovered numbers as newly verified confirmatory results.

Fresh C is a held-out, matched image-space study on 300 frozen object identifiers, independent of the original 24-object schedule-selection probe and the original 300-object evaluation cohort. The primary four-condition run (No Adapter, GFL, GFH, and C3) and the same-cohort Generic Linear/LLH addendum use the same frozen adapter checkpoint (identity recorded in Supplementary Section S8), runner source (identity recorded in Supplementary Section S8), 300-object cohort, reference/geometry inputs, target views, metric implementation, and per-object seed policy 42 + `object_idx`. The verified gates cover 1,200/1,200 primary rows and 600/600 addendum rows. The registered primary endpoint is paired C3–GFL FG-PSNR. Fresh C's 300 UIDs have zero overlap with the original evaluation cohort (including its probe), the original 1,118-object training list, and the clean-v2 evaluation cohort. The Fresh C comparison uses one frozen checkpoint and common execution protocol; hash identity with the historical Table 4 checkpoint is not used as a validity condition. The historical distinction remains: 01549 names a v1 checkpoint for its main tables and a separate v2 strong-residual, per-layer-capped adapter for Section 4.7/FAC; Fresh C is a controlled retraining, and its estimate is not presented as a numerical reproduction of the old +0.96 dB contrast. The separate post-inspection strict-276 analyses use the clean-v2 object list and record the same checkpoint SHA as Fresh C; their cohort and run provenance differ from Fresh C, and they are reported only as bounded supporting diagnostics.

#### *4.1.2. Evaluation sets and metrics*

The submitted paper describes a 50-object, 13-scale sweep; a purposive 26-object texture audit; a 24-object, 17-variant development probe; and a pooled 300-object evaluation. We retain the 50-object sweep and the 24-object probe as historical diagnostics because they motivate the scale-allocation question. Their object-level raw outputs, full selection provenance, or metric implementation records are incomplete in the current archive. The 26-object audit's object IDs, selection trace, and proxy definitions also cannot be reconstructed; its prevalence and texture-loss claims are withdrawn.

Fresh C adds a revision-era 300-object image-space comparison with foreground PSNR, LPIPS, SSIM, and Edge-SSIM. These measure agreement with target-view renderings on their named endpoints; they do not measure texture fidelity, material appearance, seams, or human preference. The separately reported 01549 preference study is a subjective choice task, not a texture-fidelity measure. Laplacian variance, RGB standard deviation, and gradient magnitude are treated only as variation diagnostics, since larger values can also reflect artifacts. Object is the inferential unit; views are averaged within object and are not independent samples.

#### *4.1.3. Implementation and dataset details*

The authenticated 01549 source names `geotex_refattn_v1/geotex_step_0002000.pt` for its main tables but provides no checkpoint hash. Fresh C uses a controlled-retraining GeoTex-Adapter checkpoint, trained from the same 1,118-object UID list (identity recorded in Supplementary Section S8). The primary run and same-cohort addendum share this frozen checkpoint, runner source, seed policy, inputs, and endpoint implementation. Fresh C uses the MVPainter-style joint six-view latent UNet and VAE, EulerDiscreteScheduler for 50 steps, six fixed target views at 256×256, and per-object seed 42 + `object_idx`, shared across all compared conditions for that object. The 300-object manifest, per-condition row counts, and input/output hashes are indexed in the Supplementary reproduction map. The frozen generator uses a conditional-only denoising prediction, without classifier-free score combination, and cached global image embeddings. The complete base pipeline instead defaults to 512-pixel views, online image embeddings, and classifier-free guidance of 2.0; the model validation call uses 75 steps. These are different sampling protocols. Our within-protocol allocation comparisons do not demonstrate equivalence to that complete default pipeline or establish which protocol difference causes the observed appearance failures (Supplementary Section S20).

Fresh C's native wrapper caps are deep 3.0, middle 3.5, and shallow 0.8. The recorded scale request, applied value, and post-scale residual norm are separate fields. The requested C3 profile is (1.25, 2.50, 1.25); its shallow value is clipped to 0.8 in this run. The Generic Linear and LLH addendum conditions are separately defined and do not equalize realized residual dose. Detailed step- and layer-group records are supplied in the Supplement.

FG-PSNR, FG-LPIPS, FG-SSIM, and Edge-SSIM are computed per object after averaging the six target views. The registered primary comparison is C3–GFL FG-PSNR; secondary and addendum tests follow their locked comparison families as documented in Supplementary Section S2. Fresh C's primary endpoint is foreground-masked PSNR. The historical Table 4 implementation and the Fresh C endpoint are not claimed to be

**Table 1. Historical 01549 report of a 13-scale sweep on the stated 50-object set. Values are reproduced from the submission and have not been independently recomputed from row-level outputs; they are retained as development diagnostics.**

| Scale | ΔFG-SSIM↑ | POS | ΔCrop-SSIM↑ | ΔPSNR↑ |
|---|---|---|---|---|
| 0.75 | −0.013 | 20/50 | +0.113 | 7.53 |
| 0.90 | +0.024 | 30/50 | +0.137 | 8.32 |
| 1.00 | +0.047 | 34/50 | +0.147 | 8.79 |
| 1.10 | +0.070 | 36/50 | +0.159 | 9.08 |
| 1.15 | +0.071 | 36/50 | +0.157 | 9.25 |
| 1.25 | +0.088 | 37/50 | +0.161 | 9.37 |
| 1.35 | +0.100 | 39/50 | +0.163 | 9.63 |
| 1.50 | +0.113 | 40/50 | +0.163 | 9.76 |
| 1.60 | +0.120 | 42/50 | +0.166 | 9.83 |
| 1.75 | +0.127 | 43/50 | +0.173 | 9.97 |
| 2.00 | +0.134 | 44/50 | +0.169 | 9.96 |
| 2.25 | +0.142 | 43/50 | +0.175 | 9.91 |
| 2.50 | +0.145 | 44/50 | +0.165 | 9.91 |

numerically identical; the Fresh C estimate is a new matched held-out result, not a reconstruction of the old +0.96 dB contrast. The original 50-object sweep and 24-object probe used separate historical records whose complete current row-level lineage is unavailable, so their implementation details are not inferred from Fresh C.

*Statistical reporting.* Fresh C reports the object-level paired mean difference and percentile-bootstrap 95% interval from 10,000 object resamples. The registered primary C3–GFL FG-PSNR contrast is unadjusted as the single primary endpoint. Secondary and addendum families and their multiplicity adjustments are detailed in Supplementary Section S2. CIs outside that prespecified family are pointwise and descriptive. The earlier three-condition 3AFC study is reported separately below as a subjective preference result; it is not a Fresh C endpoint or evidence of texture fidelity.

### 4.2. Historical Fixed-Scale Image-Space Diagnostics

Table 1 reproduces the 13-scale sweep reported in the submitted 01549 manuscript. We retain it to preserve the original development question, but the object-level outputs, full run manifest, and metric implementation mapping are not available in the current authority. Its values and Figures 2–3 are therefore historical diagnostics, not a new reanalysis or confirmatory evidence.

The submitted table reports a rising ΔFG-SSIM over most of the listed scale range and a near-plateau in the crop metric at higher settings. We retain this reported pattern only as historical motivation: absent row-level outputs and a complete run manifest, we do not treat it as an independently confirmed scale-response law. It motivates asking whether a single aggregate structural score fully describes the effect of adapter scaling.

The submitted heatmap depicts object-level variation across the reported scales, but its underlying rows cannot be reconstructed from the current evidence archive. We retain the visual diagnostic as part of the 01549 research history and do not infer a population-level response distribution from it. The updated

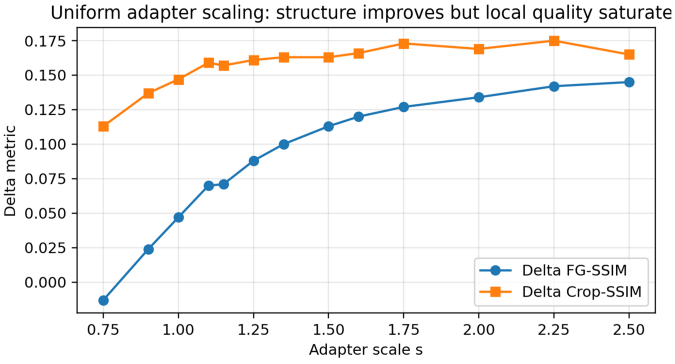



**Fig. 2. Historical fixed-scale diagnostic from the submitted 01549 paper. The plotted source reports FG-SSIM and Crop-SSIM over adapter scales; the underlying object-level run records are not available for independent reanalysis. It motivates measuring more than one endpoint, but is not a current confirmatory result.**

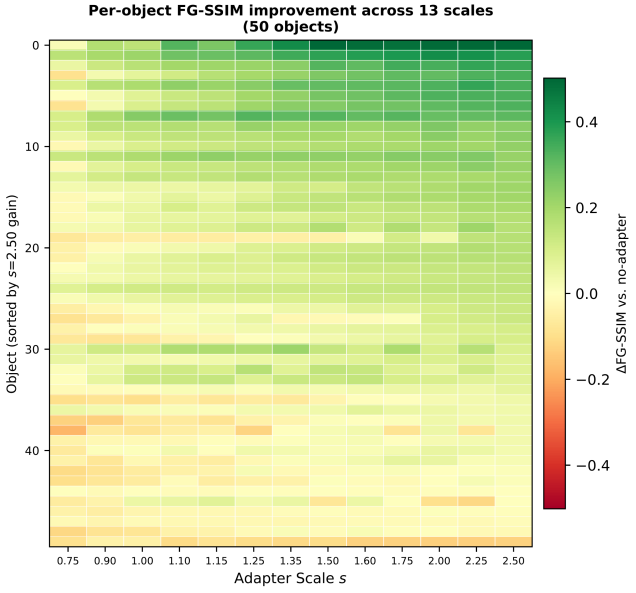



**Fig. 3. Historical per-object ΔFG-SSIM heatmap reproduced from the 01549 submission (50 objects, 13 scales). The plot illustrates the originally reported response variation; row-level provenance is incomplete, so it is not used for a current population inference.**

Fresh C analysis below uses its own frozen cohort and matched inputs.

### 4.3. C3 Development Record and Current Evaluation

The submitted paper records 17 uniform, layer-wise, and timestep-conditioned variants on a 24-object development probe and identifies C3 as its chosen profile. This study question remains relevant: does a stage-dependent request offer a useful control setting beyond a single global scale? Because the complete historical row-level data and selection provenance are not available in the current authority, the original Table 2 and Figure 4 are retained as historical development records, not as a verified ranking or proof of optimality. Fresh C evaluates the recorded C3 profile without another schedule search.

**Table 2. Representative values reported for the historical 24-object, 17-variant development probe. The full candidate set is shown in Figure 4. The underlying row-level outputs and selection-time provenance are not available for independent reanalysis; the table records the submitted development result and does not establish an optimum.**

| Variant | Reported category | ΔFG-SSIM | ΔLap Var |
|---|---|---|---|
| C4 | Late-stage high | 0.105 | −0.0120 |
| C2 | Timestep schedule | 0.104 | −0.0110 |
| B2 | Layer-wise schedule | 0.090 | −0.0135 |
| A_s2.00 | Uniform | 0.087 | −0.0062 |
| C3 | Mid-stage high | 0.084 | −0.0006 |
| A_s2.25 | Uniform | 0.079 | −0.0096 |
| A_s2.50 | Uniform | 0.078 | −0.0108 |
| B6 | Layer-wise schedule | 0.000 | −0.0027 |

The original table and plot use shape and texture-variation proxies to describe the development choice. They do not establish that texture was faithful to the reference, nor that C3 was the unique best candidate. In this revision, C3 is therefore described as the historical profile carried forward for matched evaluation, not as a theorem-derived or globally optimal schedule.

The requested profile is piecewise constant over normalized sampling progress $p$:

$$s_{\mathrm{req}}(p) = \begin{cases} 1.25, & 0 \le p < 1/3, \\ 2.50, & 1/3 \le p < 2/3, \\ 1.25, & 2/3 \le p \le 1, \end{cases} \qquad \mathrm{C3} = (1.25, 2.50, 1.25). \tag{7}$$

This is an inference-time instruction on the existing adapter path; no weights are trained. Its tested value is task- and implementation-specific. The schedule operation itself is not claimed as a new general primitive.

The source Figure 5 (26-object texture ratios) is not included as an active result because its object IDs, selection trace, and metric lineage are incomplete. The original 24-object Table 3 is moved to Supplementary Section S10 as a verbatim historical diagnostic, with no texture-fidelity or population inference. Both source assets remain preserved in the authenticated submission and forensic archive; neither decision removes the fixed-scale research question or the development record of C3.

### *4.4. Large-Scale Matched Image-Space Evaluation*

Fresh C is a revision-era matched image-space study on $N = 300$ objects. Its primary run contains No Adapter, global fixed-low (GFL), global fixed-high (GFH), and the historical C3 profile. A separate addendum evaluates Generic Linear and a layer-conditioned low–low–high temporal profile (LLH) on the same 300 object identifiers; it is not an independent cohort. All conditions within a run share the object, reference/target inputs, six target views, initial latents, and seed. Object is the inferential unit, with views averaged within object.

The registered primary contrast is C3 minus GFL on foreground PSNR (FG-PSNR). The point estimate is +0.52 dB (95% percentile-bootstrap CI [+0.40, +0.64]); C3 is higher on 206/300 objects (68.7%), with median paired difference +0.554514 dB. Registered secondary endpoints are reported with the multiplicity treatment and full paired statistics in Supplementary Section S2. PSNR, LPIPS, and SSIM quantify image-space agreement on their respective endpoints; they are not direct measures of texture fidelity or 3D material quality.

This primary contrast changes more than temporal placement. Across the 50 steps, the applied deep and middle scale sums are 82.5 each for C3 and 62.5 each for GFL; the shallow sum is 40 for both after the native 0.8 cap. The Fresh C estimate therefore supports the recorded C3 intervention relative to GFL under this wrapper, but cannot be attributed to timing alone or interpreted as a pure schedule-shape effect.

The C3–GFL comparison supports a bounded statement: on this frozen, UID-disjoint cohort and image-space endpoint, the recorded C3 intervention improves mean FG-PSNR relative to GFL. The paired conditions use one frozen checkpoint, common runner, seed policy, inputs, target views, metrics, and recorded native-cap semantics. The historical 01549 v1/v2 checkpoint distinction is preserved, and Fresh C is not presented as a numerical reproduction of the old Table 4 result. This historical difference does not alter the internal pairing or held-out status of Fresh C.

The counter-results delimit that finding. GFH has more favorable cohort means than C3 on all four displayed endpoints. Generic Linear also has more favorable means than C3 on all four; for C3 minus Generic Linear, the mean FG-PSNR difference is −0.593 dB (95% CI [−0.837, −0.351]), although the median is +0.073956 dB and C3 is higher on 155/300 objects (51.7%). LLH's FG-PSNR mean exceeds C3 by +0.621 dB, while its median difference is −0.153 dB and only 140/300 objects favour LLH; its lower mean FG-LPIPS favours LLH. For LLH versus Generic Linear, the FG-PSNR CI includes zero, which is not evidence of equivalence; the FG-LPIPS difference favours Generic Linear. The endpoint means and object-level distributions do not identify a universal winner.

Fresh C has no validated texture-fidelity endpoint. RGB, Laplacian, and gradient responses describe variation: a larger value can represent detail or an artifact. Figure 5 shows selected full-object/same-ROI fixed-scale and Generic Linear comparisons as qualitative context; they do not estimate prevalence, colorimetric agreement, or texture fidelity.

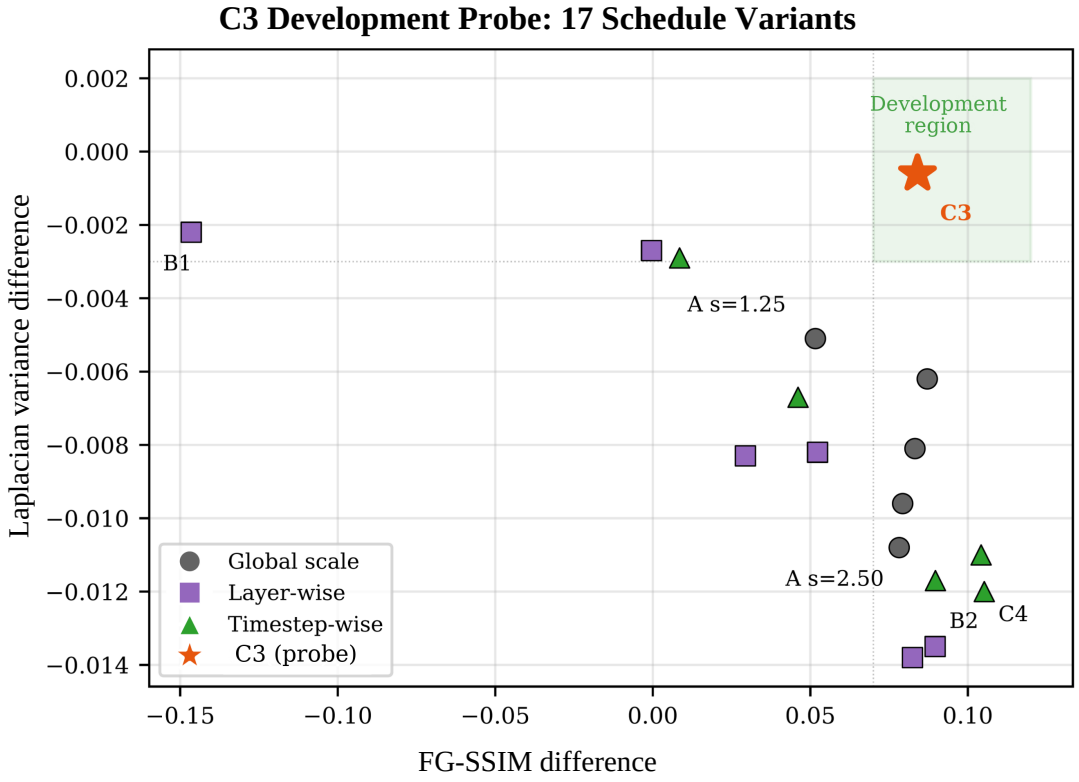



**Fig. 4. Historical 17-variant development probe from the submitted 01549 paper (24 objects). The axes show changes in FG-SSIM and Laplacian variance; variance is a texture-variation diagnostic, not a fidelity measure. "C3" identifies the selected historical profile, not an optimum or general schedule law. The candidate-level outputs are unavailable for independent reanalysis.**

**Table 3. Fresh C image-space condition means. The primary run contains No Adapter, GFL, GFH, and C3. Generic Linear and LLH are revision-era addendum conditions evaluated on the same object identifiers, not a second independent cohort. Values are means of object-level metrics over six target views.**

| Method | *N* | FG-PSNR↑ | FG-LPIPS↓ | FG-SSIM↑ | Edge-SSIM↑ |
|---|---|---|---|---|---|
| Primary run | | | | | |
| No Adapter | 300 | 6.9779 | 0.2288 | 0.4086 | 0.4314 |
| GFL | 300 | 11.3077 | 0.2051 | 0.3500 | 0.4728 |
| GFH | 300 | 12.5777 | 0.1891 | 0.4201 | 0.4961 |
| C3 (TCAS) | 300 | 11.8301 | 0.1998 | 0.3746 | 0.4773 |
| Same-cohort revision-era addendum | | | | | |
| Generic Linear | 300 | 12.4230 | 0.1862 | 0.4026 | 0.4880 |
| LLH | 300 | 12.4506 | 0.1876 | 0.3911 | 0.4863 |

**Table 4. Fresh C paired mean differences (A minus B) and pointwise 95% object-bootstrap CIs. Positive PSNR/SSIM and negative LPIPS favour A. C3–No Adapter/GFH are post hoc; the addendum shares object IDs. Full adjusted families are in Supplement S2.**

| Contrast | FG-PSNR (dB) | FG-LPIPS | FG-SSIM | Edge-SSIM |
|---|---|---|---|---|
| C3 − GFL | +0.522 [0.404, 0.641] | −0.005345 [−0.006326, −0.004403] | +0.02453 [0.02120, 0.02805] | +0.004494 [0.003411, 0.005586] |
| C3 − No Adapter | +4.852 [4.553, 5.142] | −0.02905 [−0.03352, −0.02464] | −0.03401 [−0.04214, −0.02591] | +0.04586 [0.04114, 0.05054] |
| C3 − GFH | −0.748 [−0.885, −0.609] | +0.01070 [0.008958, 0.012513] | −0.04553 [−0.04937, −0.04167] | −0.01881 [−0.02084, −0.01683] |
| C3 − Generic Linear | −0.593 [−0.837, −0.351] | +0.01351 [0.01160, 0.01549] | −0.02810 [−0.03197, −0.02439] | −0.01069 [−0.01300, −0.00850] |
| LLH − C3 | +0.621 [0.391, 0.855] | −0.01214 [−0.01386, −0.01048] | +0.01658 [0.01311, 0.02003] | +0.009033 [0.006867, 0.011296] |
| LLH − Generic Linear | +0.028 [−0.015, 0.070] | +0.001372 [0.000762, 0.002081] | −0.01152 [−0.01337, −0.00996] | −0.001660 [−0.002096, −0.001256] |

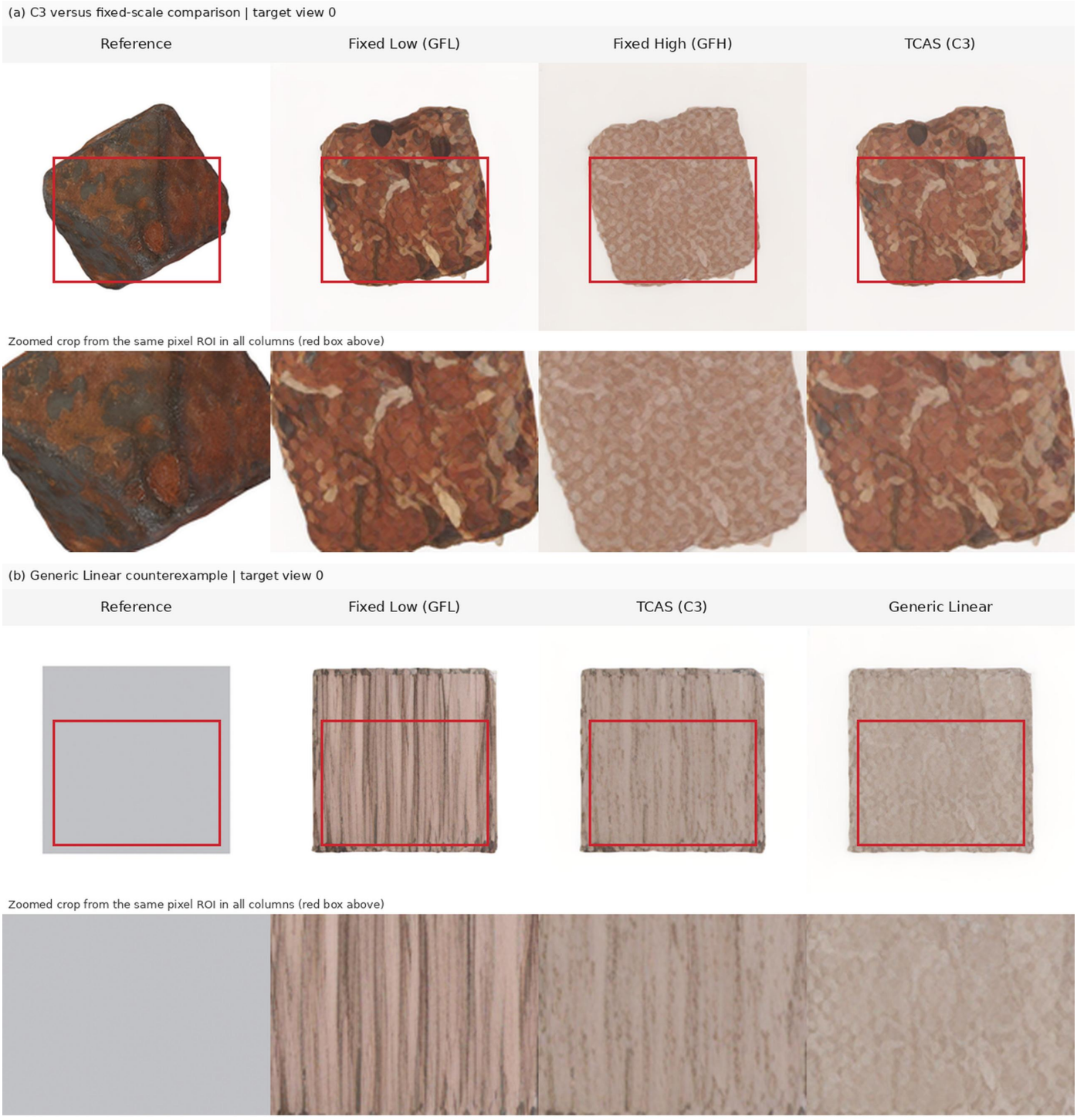



**Fig. 5. Matched Fresh C target-view RGB outputs for the same frozen object, reference, camera, checkpoint, runner, seed policy, and inference protocol. (a) shows a selected fixed-scale case with an endpoint trade-off between C3 and GFH. (b) shows a counterexample where C3 minus Generic Linear is** −3.491 **dB FG-PSNR and** +0.08172 **FG-LPIPS, favoring Generic Linear on both endpoints. Each context marks the exact pixel ROI shown below. Outcome-selected examples do not estimate prevalence or texture fidelity; the RGB crops do not establish colorimetric agreement. Full six-view panels remain in Supplementary Figure S1; model credits and image/ROI provenance are in Supplementary Table S1 and the Fresh C figure manifest.**

### 4.5. Applied-Scale Budget and Temporal Placement

Fresh C compares the recorded C3 profile with GFL under the same frozen checkpoint, runner, seed policy, inputs, and target views. Its applied deep/middle scale sums are 82.5 for C3 and 62.5 for GFL, while the shallow sum is 40 for both after the 0.8 cap. The registered C3–GFL improvement therefore combines a change in middle-third strength with a larger per-group nominal budget; it does not isolate temporal placement.

A separate post-inspection placement analysis used the clean-v2 strict-276 cohort and a common runner within that run. It is not the Fresh C cohort, and it is not an independent replication of the historical 24-object selection probe. Temporal L–L–H means low requests during early and middle steps and high requests during late steps; it uses a different shallow-group control from Fresh C's layer-conditioned LLH addendum. The latter requests (1.25,1.25,2.50) in deep/middle and (0.50,0.50,0.75) in shallow; the older global placement comparison keeps shallow capped at 0.8. Neither is identical to the new 16-step Late-high condition. With 50 steps, the strict-276 run assigns 17/16/17 steps to the three windows. C3 and temporal L–L–H have near-matched applied-scale sums (82.50 versus 83.75 in deep and middle; 40 in shallow), but their post-scale residual magnitudes are not matched.

Against the fixed-mean profile, C3's strict-276 FG-PSNR difference is +0.502875 dB (95% CI [+0.462018, +0.542244]). These results show sensitivity to the tested stage allocation, not a universally preferred placement or an equal-realized-dose effect.

Fresh B is a separate $N = 150$ cohort and reports LLH against LFM-EXACT, not C3. The paired LLH–LFM-EXACT differences are +0.325 dB FG-PSNR (95% CI [+0.209, +0.446]) and −0.00460 FG-LPIPS (95% CI [−0.00563, −0.00357]); both intervals fall within the recorded practical margins of ±0.5 dB and ±0.01. The saved lock states a pre-run freeze time, but the earliest preserved commit containing it is later than that time. We therefore report Fresh B as retrospective sensitivity rather than authenticated preregistered confirmation.

A separate Core-7 summary uses the same 276 object labels and recorded checkpoint lineage as Fresh C but does not establish common runner or complete input-tensor identity across its seven conditions. Its post-hoc C3–GFH contrasts point in opposite directions for Full-view PSNR (+0.725229 dB, 95% CI [+0.647134, +0.806467]) and FG-PSNR (−1.214235 dB, 95% CI [−1.313952, −1.114705]). It remains a retrospective diagnostic, not a same-runner confirmation or reproduction of the original +0.96 dB result (Supplementary Section S5).

### 4.6. Matched Stage Allocation at Equal Applied Budget

The six-condition matrix contains 1,800 object–condition rows, with complete input-hash and common-seed checks. It is a post-inspection addendum on Fresh C's 300 objects, not an independent holdout. Early-, middle-, and late-high allocate the same 16 high steps to updates 0–15, 17–32, and 34–49. Deep and middle use low 1.25 and high 2.50; shallow remains capped at 0.8. Fixed mean uses 1.65 before the same cap. Thus all four allocation conditions have per-group applied sums (82.5, 82.5, 40), or 205 in total. Fixed Low and Fixed High are separate strength controls and do not share that budget. Generic Linear remains a strong practical baseline in Table 3, rather than being silently treated as equal-budget.

C3 minus Fixed Mean is −0.13 dB (pointwise 95% object-bootstrap CI [−0.17, −0.10]). C3 minus Late-high is −0.56 dB (pointwise 95% object-bootstrap CI [−0.63, −0.49]). C3 minus Early-high is +0.50 dB (pointwise 95% object-bootstrap CI [+0.37, +0.62]). C3 also trails Fixed Mean and Late-high on

**Table 5. Post-inspection strict-276 placement analysis. Entries are C3 minus temporal L–L–H with pointwise 95% object-bootstrap intervals; positive PSNR/SSIM and negative LPIPS differences favor C3. Temporal L–L–H is favored on all six formal endpoints. The analysis is separate from Fresh C, and its scalar sums do not match realized residual dose.**

| Endpoint | Mean difference [95% CI] | Direction |
|---|---|---|
| Full-view PSNR (dB) | −0.268100 [−0.311101, −0.227797] | L–L–H |
| FG-PSNR (dB) | −0.177084 [−0.230539, −0.130784] | L–L–H |
| FG-SSIM | −0.008851 [−0.011455, −0.006291] | L–L–H |
| Edge-SSIM | −0.035301 [−0.036581, −0.034031] | L–L–H |
| FG-LPIPS | +0.001896 [+0.001205, +0.002578] | L–L–H |
| Full-view LPIPS | +0.008658 [+0.007465, +0.009831] | L–L–H |

**Table 6. Fully matched Fresh C control means ($N$ = 300 per condition). Timing and fixed-mean rows share exactly the same per-layer applied-scale sums. This same-cohort addendum is post-inspection.**

| Control | FG-PSNR↑ | FG-LPIPS↓ | FG-SSIM↑ | Edge-SSIM↑ | PSNR winners |
|---|---|---|---|---|---|
| GFH / Fixed High | 12.58 | 0.1891 | 0.4201 | 0.4961 | 172 |
| GFL / Fixed Low | 11.31 | 0.2051 | 0.3500 | 0.4728 | 37 |
| Fixed Mean | 11.96 | 0.1978 | 0.3864 | 0.4843 | 2 |
| Early-high | 11.33 | 0.2045 | 0.3696 | 0.4765 | 25 |
| C3 / Middle-high | 11.83 | 0.1998 | 0.3746 | 0.4773 | 0 |
| Late-high | 12.39 | 0.1951 | 0.3788 | 0.4914 | 64 |

**Table 7. Object-cluster tests of the observed layer-by-window response maps. A2 is exploratory; A3b is retrospective Fresh B sensitivity because the preserved history cannot authenticate pre-run availability of its lock. Interaction share and cell RMSE summarize the observed map, not a causal effect.**

| Map, cohort | Endpoint | Wald $W(8)$ | Reported $p$ | Share; cell RMSE |
|---|---|---|---|---|
| A2, $N$ = 300 | FG-PSNR | 1879.963 | $< 10^{-300}$ | 56.8%; 0.1222 dB |
| A2, $N$ = 300 | FG-LPIPS | 481.081 | $8.04 \times 10^{-99}$ | 43.5%; 0.001695 |
| A3b, $N$ = 150 | FG-PSNR | 525.700 | $3.43 \times 10^{-107}$ | 36.0%; 0.0205 dB |
| A3b, $N$ = 150 | FG-LPIPS | 225.015 | $5.37 \times 10^{-43}$ | 14.0%; 0.000342 |

**Table 8. Selected paired results from the failure-enriched 26-object color diagnosis. Negative CIEDE2000 differences indicate lower reference-color error for the first condition. The set is descriptive and does not estimate population prevalence.**

| Contrast | Mean CIEDE2000 difference [95% CI] | Objects |
|---|---|---|
| GFL – No Adapter | −19.654 [−26.370, −13.248] | 24/26 lower |
| LLH – GFL | +3.583 [+0.462, +6.712] | – |

FG-LPIPS, FG-SSIM and Edge-SSIM. Complete paired endpoint distributions, 50,000 object-bootstrap resamples and post hoc Holm-adjusted contrasts are supplied in Supplementary Section S16. The rank count is specific to FG-PSNR and these six conditions; it does not establish a universal quality winner. We retain C3's development identity but reject a generally optimal middle window. Residual amplitudes differ along the resulting trajectories: this restricts a further mechanism claim, while the equal-budget allocation comparison still measures the total effect of stage allocation.

### *4.7. Layer-by-Window Sensitivity*

The A2 and A3b maps test response heterogeneity over three adapter groups and five denoising windows. Their object-cluster statistics describe variation within the tested interventions; they are not effect estimates for a dose-independent schedule mechanism.

The earlier A3b common-support calculation compared 10-step integrated raw $L_2$ norms across three layer groups. The saved corrections have different spatial resolutions (24×16, 48×32, and 96×64), and the traces omit channel counts, so absolute norms are dimension-dependent across groups. We withdraw the reported 0/2,250 overlap as a cross-layer dose-support conclusion. Together with the Fresh B lock chronology, A3b remains retrospective sensitivity evidence. Its map shows response heterogeneity under the tested interventions, but does not identify a dose-independent layer-by-window effect or an optimal schedule.

A separate retrospective analysis on the `FRESH_CONFIRM_300` cohort found nine of ten plus-one-corrected associations between LLH–GFL image-space deltas and ground-truth descriptors; the FG-LPIPS/coverage test had adjusted $p$ = 0.0508. FG-PSNR means ranged from +4.140 dB in the lowest Laplacian-variance quartile to −0.087 and −0.029 dB in the upper two quartiles. These are cohort-specific associations, not a causal complexity effect, a threshold, a schedule selector, or a texture-fidelity result.

### *4.8. Appearance and Color Failure Analysis*

The selected color diagnosis contains 26 failure-enriched objects: six earlier figure cases and 20 Fresh C panels. It is not a random sample, a new holdout, or a population estimate. No Adapter, GFL, and LLH were compared on the same inputs, checkpoint, and runner with six target views per object (78 object-condition records). The following paired object-bootstrap intervals are descriptive (20,000 resamples; seed 61008), and no confirmatory multiplicity family was defined.

GFL lowered reference-relative CIEDE2000 in 24/26 objects, while visible pink or purple shifts remained in two legacy Fig. 4 cases under No Adapter, GFL, and LLH. Their mapping to the original submitted panels cannot be recovered; the traceable Fresh C example in Figure 6 is a different object and is not presented as a reproduction of those cases. The decoded RGB output showed the shifts before UV projection or baking, and a late denoising snapshot (update 40 of 50) showed one of them. In that case, a ground-truth VAE encode–decode isolate had

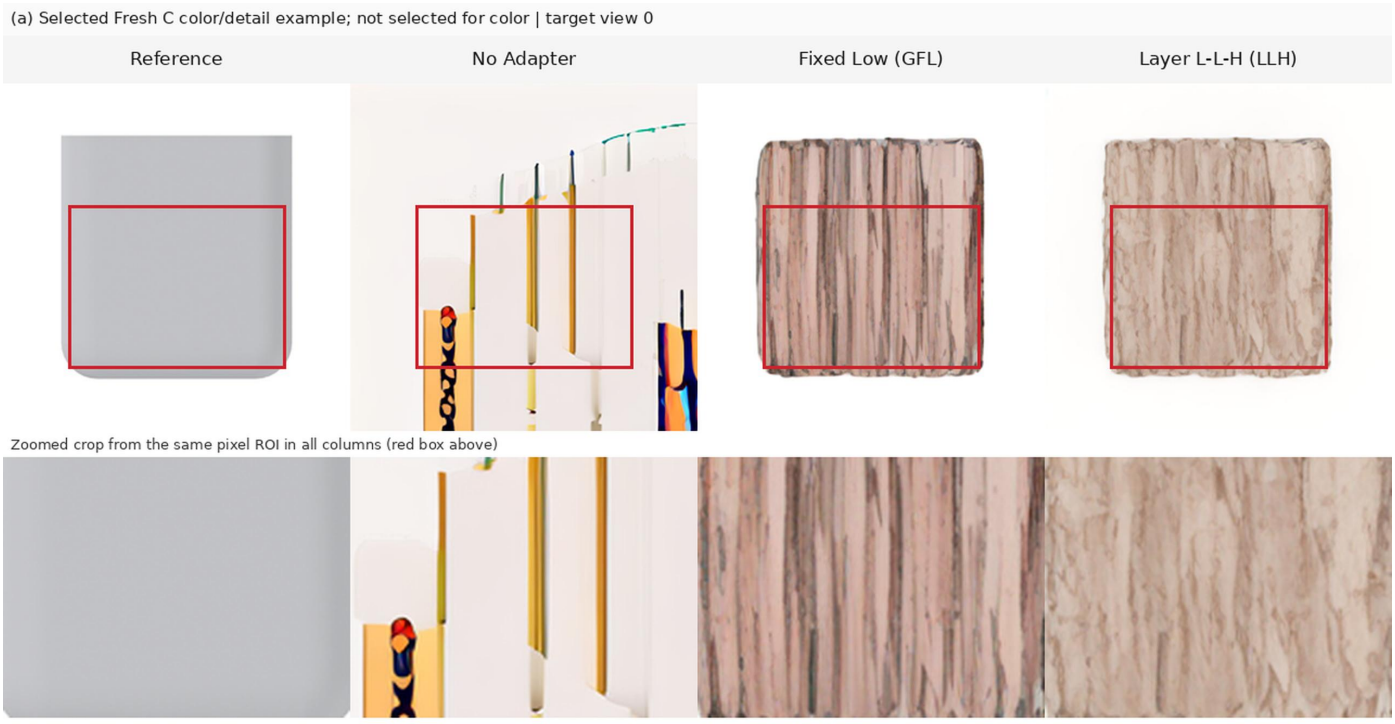



**Fig. 6. Matched Fresh C target-view RGB outputs for UID `f6979d7ff8e54c4daf00fee46f09a139`: reference, No Adapter, GFL, and LLH. Each column shows the same target view (view 0), full context, and shared pixel ROI. This previously viewed object is one of the 20 Fresh C panels in the 26-object diagnosis; its frozen stratum was the lowest C3–GFL FG-LPIPS, not color-selected. It is not independent confirmation or a population estimate, and it is distinct from the unrecoverable legacy 01549 color panels. Full attribution and source/ROI hashes are listed in Supplementary Table S1 and the figure manifest.**

CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation in that case; it does not identify the complete cause or rule out other model and conditioning effects.

A locked P0 trace reproduced the archived RGB pixels for three preselected objects in repeated Python 3.13.5 runs; the Python 3.10 trace reproduced its corresponding archived observer. The first cross-runtime numerical divergence occurred after multiplying the identical FP16 initial latent by the scheduler's initial noise scale (maximum absolute difference 0.0625; RMSE 0.00966). This localizes an execution-stack difference, not the color-generation cause; the three-object trace does not establish cohort-wide cross-stack equivalence.

LLH–GFL also increased mean CIEDE2000 while reducing foreground gradient magnitude (−0.06772), high-frequency energy (−0.00887), and color entropy (−0.598). Lower local variation therefore did not imply better reference-color agreement or greater texture fidelity. The terminology follows Section 3.2: texture variation, image-space agreement, reference-color agreement, subjective preference, baked 3D appearance, and PBR/material fidelity are reported separately.

Three tested interventions failed to validate a color correction. The C1 chroma-anchor candidate passed its four-object numeric gate (mean unseen-view ΔFG-CIEDE2000 = −1.5602, 95% CI [−2.2621, −0.6299]) but failed locked visual review, including a cyan/blue rim. B2's four-object numeric result was +0.1729 (95% CI [−0.2044, +0.6810]), with only one of four objects improving and no clear visual correction. A feature-relative residual gate increased mean CIEDE2000 versus C3 by 1.35 across 26 objects and by 3.18 among the six earlier cases, with no clear visual reduction in the pink shift. These are sensitivity and failed-candidate results; no color-cast repair is validated.

The historical $N = 20$ GLB-native check used 20 UV-compatible objects from the original 24-object pool and 11 unseen views, rendered with unlit base color. C3–GFL FG-PSNR was −0.066 dB (95% CI [−0.413, +0.290]), FG-LPIPS was −0.004719 (95% CI [−0.007504, −0.002544]), and CIEDE2000 was +0.133 (95% CI [−1.161, +1.319]): endpoint directions are mixed, and these results do not establish full PBR/material fidelity or seam quality. The earlier Fresh C 24-object attempt failed its input gate and is retained as an implementation diagnostic. The five-object bake has now been rerun with corrected pole masks, scalar-alpha decoding and resolution-matched GT-256 inputs. Its descriptive C3–GFH PSNR difference is +2.37 dB, with lower SSIM; Late-high has higher mean PSNR and lower LPIPS than C3. This earlier pilot is not pooled with the new ten-object study in Section 4.9, and does not independently validate a universal C3 benefit. Separate REPEAT-addressed results also required a bilinear-weight correction. The historical $N = 20$ study, this five-object archive, and the newly frozen technical cohort are not pooled. The stored GeoTex condition named native_default_1x is a Constant 1.0 request under native caps: its shallow correction is multiplied by 0.8, whereas the unset-scale wrapper applies the raw

correction. An execution check of the actual wrapper branches rejects default-path equivalence. This condition is therefore labeled Constant 1.0 (capped), not an unmodified pipeline.

*Fresh C full-object and same-ROI examples.*. The submitted 26-object texture audit and its quantitative Figure 5 cannot be re-traced to object IDs, selection criteria, raw values, and an executable metric definition in the current evidence archive. We therefore remove its 92% prevalence statement, its "true shape gain" classification, and its RGB-standard-deviation/gradient ratios from the scientific claims. Those statistics would describe variation, not texture fidelity, even if their source records were complete.

Figure 7 shows Fresh C full-object appearance beside the same-pixel ROI for a selected C3–GFL favorable case and an adverse case. Figure 5 shows the fixed-scale comparison and a Generic Linear counterexample. These outcome-selected RGB illustrations do not estimate prevalence or texture fidelity. The full six-view, seven-condition panels, including No Adapter and LLH, remain in Supplementary Figure S1.

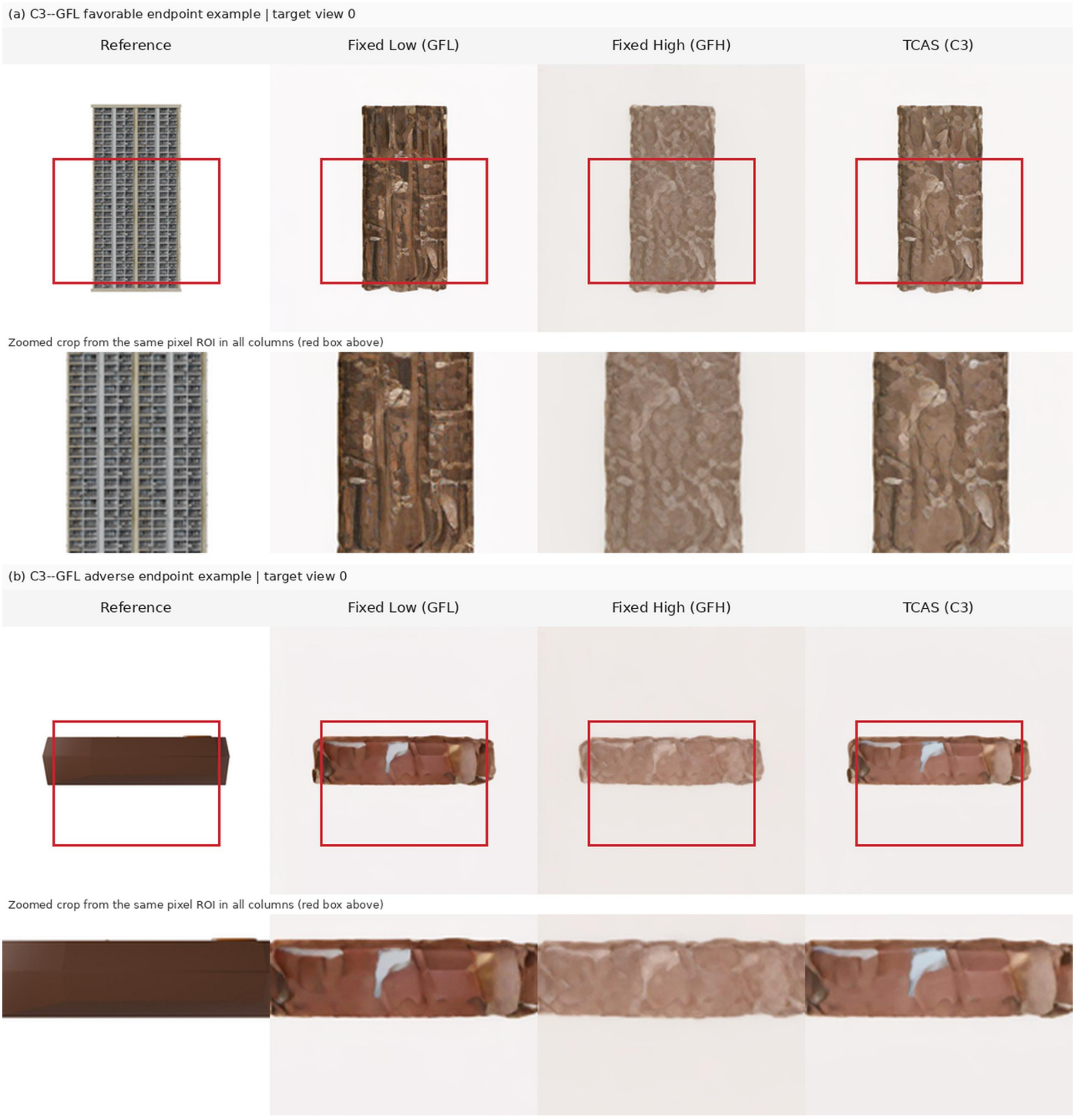



**Fig. 7. Matched Fresh C target-view RGB outputs for the same frozen object, reference, camera, checkpoint, runner, seed policy, and inference protocol. Each column pairs the full context (shared ROI in red) with its exact pixel crop. Case (a) was selected because C3 favors GFL on paired FG-PSNR and FG-LPIPS; both endpoints favor GFL in case (b). These outcome-selected examples do not estimate prevalence, texture fidelity, reference-color agreement, or downstream 3D/PBR quality. Model credits are in Supplementary Table S1; image hashes and ROI coordinates are in the Fresh C figure manifest.**

### *4.9. Calibrated Baked Appearance on New Objects*

We froze ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle-count and camera-mask alignment criteria. Of the 300 screened inputs, only ten passed all criteria before the cutoff; the planned maximum was twenty. The ten are not new to the GeoTex image domain, but their baked outcomes had not been inspected during selection. This is a compatibility-selected pilot, not a representative twenty-object confirmation. Five earlier bake objects are reported separately.

Seven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel source resolu-

tion, atlas resolution, hole filling and eleven unseen evaluation cameras. Original geometry, UVs and legal base-color material semantics are retained and checked. Unseen cameras do not enter the bake. GT-256 uses the same source resolution, masks and bake parameters and has exactly the same raw atlas coverage as every generated condition. Native REPEAT sampling passes boundary and integer-period tests; pole-view RGB orientation is corrected without rotating raw-camera masks. These calibrations control projection artifacts; evaluation is unlit base color, not PBR fidelity. Supplementary Section S18 records the checks and object-level data.

**Table 9. New ten-object unlit unseen-view bake pilot. Eleven views are averaged per object. Higher PSNR/SSIM and lower LPIPS/CIEDE2000 are favorable. Constant 1.0 is an explicitly capped request, not the unset-scale pipeline. All seven original generated conditions and all ten eligible objects are retained; the two subsequently added controls marked † are exploratory; GT-256 is a separate matched reference bake, not a theoretical bound.**

| Condition | FG-PSNR | FG-LPIPS | FG-SSIM | CIEDE2000 |
|---|---|---|---|---|
| GT-256 reference bake | 22.38 | 0.0590 | 0.7683 | 4.44 |
| Native residual† | 11.13 | 0.1372 | 0.3376 | 26.47 |
| No Adapter† | 5.92 | 0.1600 | 0.3829 | 40.92 |
| Constant 1.0 (capped) | 11.93 | 0.1354 | 0.3504 | 24.18 |
| Fixed Low | 12.48 | 0.1317 | 0.3835 | 22.71 |
| Fixed High | 12.29 | 0.1305 | 0.4587 | 23.27 |
| Fixed Mean | 12.62 | 0.1303 | 0.4193 | 22.32 |
| C3 / Middle-high | 12.48 | 0.1302 | 0.4059 | 22.16 |
| Late-high | 12.92 | 0.1277 | 0.4259 | 21.65 |
| Generic Linear | 11.60 | 0.1276 | 0.4458 | 24.71 |

The internally predeclared primary contrast was Late-high minus Fixed Mean on unseen-view FG-LPIPS, proposed from the earlier pilot and tested on the ten newly selected bake objects. Its mean is −0.00265 (95% object-bootstrap CI [−0.00414, −0.00137]; exact paired sign-flip $p = 0.00195$), with all ten objects favorable. This is approximately a 2.0% reduction relative to Fixed Mean's mean LPIPS, and identifies a small allocation benefit within this sampling protocol. Generic Linear has a slightly lower mean LPIPS but lower PSNR; Late-high does not dominate every endpoint or control. The matched GT-256 reference bake has mean LPIPS 0.0590, versus 0.1303 for Fixed Mean and 0.1277 for Late-high. A post hoc description therefore puts the absolute gain in context: it closes only 3.7% of the mean Fixed Mean–GT reference gap (pointwise object-bootstrap CI [2.3%,5.3%]). This ratio of paired means is not a theoretical ceiling, a new primary test or evidence of acceptable material fidelity. Leaving out each object in turn keeps the mean Late–Mean difference between −0.00294 and −0.00208, but all substantial reference-appearance discrepancies remain. Supplementary Section S19 supplies the paired values and calculation.

C3–Fixed High has mean PSNR +0.19 dB (95% CI [−0.65, +0.87]) and LPIPS −0.00036 ([−0.00396, +0.00336]); neither separates under the four-comparison secondary Holm family (both adjusted $p = 1$). Its SSIM is lower by 0.05274 ([−0.07459, −0.03170]; Holm $p = 0.00781$). Thus the old five-object C3 PSNR advantage does not repeat stably on this new pilot. We retain C3 as the historical candidate and do not rename Late-high as C3.

Across the seven conditions, mean within-object Spearman correlations between generated-view and baked unseen-view rankings are 0.846 for PSNR, 0.804 for LPIPS, 0.746 for SSIM and 0.907 for Edge-SSIM. Broad rankings are positively associated, while the winning condition changes on 7/10 objects for LPIPS (1/10 for PSNR). These are different statements: we do not claim a general reversal of rankings. Full-object Figure 8 also shows persistent appearance errors, including the Tire's brown output against a gray reference. A lower LPIPS does not establish faithful material reproduction.

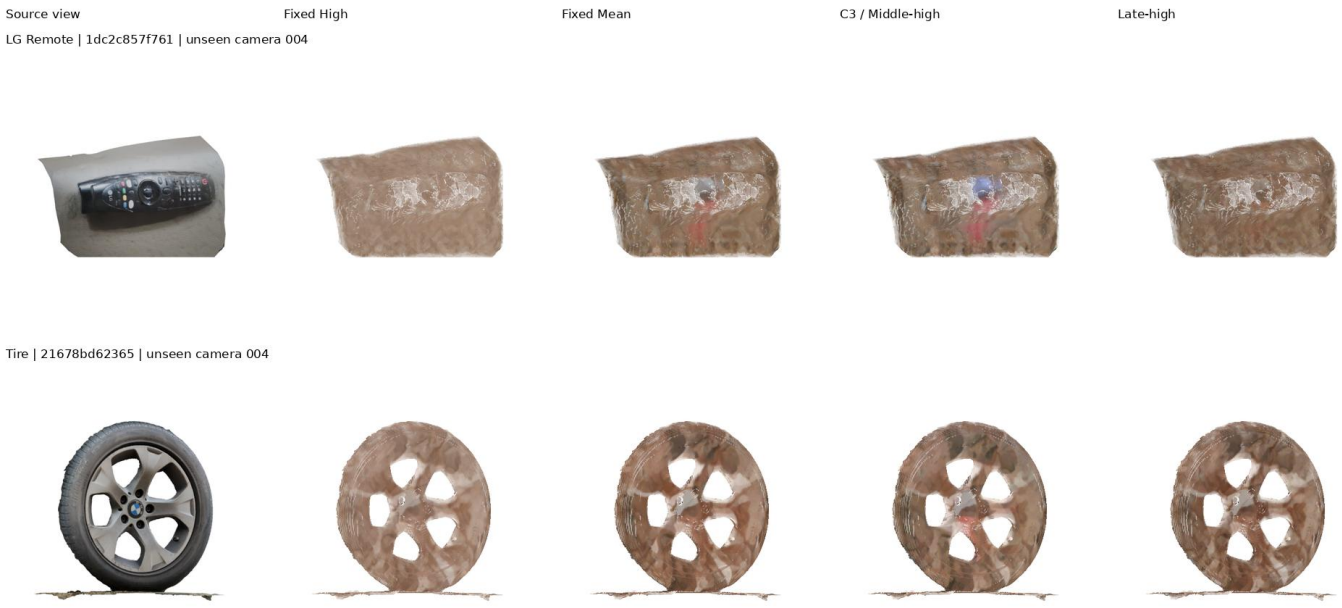



**Fig. 8. Actual source and unlit baked outputs at the same unseen camera, with whole-object framing and no color or detail editing. These are the first two input-order objects with verified publication licenses, not outcome-selected examples. All seven conditions plus source and GT-256 bake for seven rights-cleared objects appear in Supplementary Figure S2; all ten objects enter the numerical analysis. LG Remote: mohd.akmal.mahyudin, CC Attribution; Tire: Koto3D Stephane Vogley, CC Attribution (full source links in Supplementary Section S18). The Tire illustrates a retained appearance failure.**

*Exploratory native residual controls.*. We subsequently generated the unset-scale native residual branch and No Adapter on the same ten objects, using the frozen 50-step protocol. A regenerated Constant 1.0 (capped) control matches the earlier RGB outputs bitwise; both new controls preserve the original mesh and matched GT coverage, and repeated GT metrics are identical. Late-high minus Native residual has PSNR +1.79 dB (pointwise 95% object-bootstrap CI $[+0.46, +3.06]$) and LPIPS −0.00951 ($[-0.01772, -0.00263]$). C3 minus Native residual has PSNR +1.35 dB ($[+0.13, +2.56]$) and LPIPS −0.00699 ($[-0.01326, -0.00159]$). Both schedules have more favorable means on all five endpoints, but none of their five contrasts separates under the exploratory 15-test Holm family (these PSNR/LPIPS contrasts each have adjusted $p = 0.297$). This is neither independent confirmation nor equivalence evidence. No Adapter has worse mean PSNR, LPIPS, Edge-SSIM and color agreement but higher SSIM than Native residual; only its Edge-SSIM contrast separates after correction (−0.218, adjusted $p = 0.0293$). Full contrasts and unchanged failed examples are in S20. The complete default-pipeline protocol differs in resolution, image-feature preparation and classifier-free guidance, so these controls establish only within-protocol behavior.

*Complete base-pipeline diagnostic.*. On the two previously displayed failures, we separately loaded the same native base assets without GeoTex and called the full pipeline at its default 50 steps and a 75-step sensitivity setting. All four calls complete, with 512-pixel views, online image features, classifier-free guidance 2.0 and the native Euler ancestral scheduler. The original RGBA conditioning source is used because the API rejects RGB input; its preprocessing also modifies some bright foreground pixels. Conditioning-source correspondence and paired initial latent identities are recorded. The front-looking outputs visibly retain the remote and tire's reference colors/details, unlike the controlled generator's failures, although other views remain imperfect. These protocols differ on several axes, and their camera conventions are not verified as identical. We therefore report complete grids in S21, without cross-protocol paired GT metrics, causal attribution or a population quality claim. This diagnostic motivates separating sampling-path differences before further schedule or algorithm development; it does not validate GeoTex deployment or resolve novelty.

### *4.10. Cross-Interface Evaluation*

Table 10 places the primary GeoTex path beside the auxiliary MV-Adapter and MVDiffusion interventions. The methods use different conditioning interfaces and scale semantics; object-level results are not pooled across systems.

Under the raw A-minus-B convention, L–LLH has higher CIEDE2000 (worse reference-color agreement) than G–FL, lower PSNR/FG-SSIM/Edge-SSIM, lower GT-relative error against the recorded variation statistics, and no separated FG-LPIPS difference. The lower variation-statistic error is not evidence of texture fidelity. The earlier negative CIEDE2000 value was an improvement-coded effect, not a raw contrast; the direction audit and all five comparisons are reported in Supplementary Table S12.

For the 98-object MV-Adapter probe, the original additive-residual bootstrap had an invalid zero-power null and was not used for interaction inference. The exploratory object-cluster

**Table 10. Interface-specific intervention evidence. Nominal budgets are described within each implementation; no cross-interface realized-dose match is available. These rows are not architecture-matched replications. The MVDiffusion contrast is reported as raw condition A minus condition B; lower FG-LPIPS, CIEDE2000, and GT-relative texture error are favorable, whereas higher PSNR, FG-SSIM, and Edge-SSIM are favorable.**

| System and injected signal | Control axis and budget | Cohort and principal result | Evidence grade and limitation |
|---|---|---|---|
| GeoTex-Adapter, Fresh C | Five-channel normal/depth/foreground adapter residuals added to UNet ResNet features; temporal requests with native layer caps. C3–GFL applied deep/middle sums differ (82.5 vs. 62.5); shallow sums are both 40. | $N$ = 300. C3–GFL FG-PSNR is +0.522 dB [0.404, 0.641], but GFH and Generic Linear have more favorable means than C3 on all four displayed endpoints. | Matched, same-runner Fresh C image-space comparison; identifies the named paired contrast, not timing alone, a schedule optimum, or baked material quality. |
| MV-Adapter, layer/window probe | Condition guider plus decoupled multi-view attention; one of four down-block points is raised from 0.75 to 1.00 in one of five 10-step windows. Group residual-dose ratio is 1:0.57:1.46; no equal-realized-dose map was run. | 98/99 planned objects. The exploratory four-metric Holm-corrected cluster-Wald family found no interaction detected after correction (primary FG-LPIPS Holm $p$ = 0.303). | Interface-specific layer/window sensitivity; not C3 temporal transfer. A corrected non-detection does not show no interaction or equivalence. |
| MV-Adapter, layer/time profile panel | Four condition-guider features were scaled per point; L-FIX, L-LHL, and L-LLH add 228 outputs to the existing global schedules on the same 76-object Exact holdout. | Layer-wise/global contrasts are metric-dependent. The two within-profile temporal contrasts show no consistent pattern; their 30 pointwise intervals are unadjusted. | Retrospective: protocol and outputs first appear in one preserved commit. This feature-scaling interface differs from GeoTex residual correction and does not establish TCAS transfer. |
| MVDiffusion, compatibility interface | Correspondence-aware decoder-side CP-Blocks; the tested multiplier is a product of a layer factor and temporal scale over five blocks. The 75-object panel used the compatibility base; a later check found 1,552/1,552 loaded tensors and 12/12 views identical to the native mirror on one object. | $N$ = 75, 12-view output grids, six frozen target views scored per object, and 50 steps. For raw L–LLH minus G–FL means: PSNR −0.4538, FG-SSIM −0.01939, Edge-SSIM −0.005941, FG-LPIPS +0.002745, CIEDE2000 +1.3730, and GT-relative texture error −0.08229. Full pointwise 95% CIs for all endpoints and five contrasts are in Supplementary Table S12. | Within-interface, mixed endpoint comparison on a different conditioning path; the 75-object panel itself was not rerun on the mirror. No shared scale semantics, cross-system dose match, or architecture-causal inference. |

Wald test is reported with its corrected family; its non-detection does not establish absence of interaction. In the separate 76-object Exact-holdout panel, layer-wise versus global contrasts trade PSNR, color, and ground-truth-relative texture endpoints against FG- and Edge-SSIM. For temporal placement, only one of six pointwise 95% intervals excludes zero in L-LLH versus L-FIX (direction-adjusted CIEDE2000 +0.072790 [+0.004371, +0.156696]); only one does so in G-LLH versus G-LHL (Edge-SSIM −0.000261 [−0.000522, −0.000032]). The 30 intervals are descriptive and unadjusted. The protocol and outputs first appear in the same preserved commit, so this is retrospective sensitivity evidence, not confirmation or an equivalence test. The MVDiffusion panel used a compatibility deployment; a separate check matched its loaded tensors and one object's rendered views to a native mirror, but the full panel was not rerun. These probes remain interface-specific and do not establish architecture-invariant scheduling or transfer of the GeoTex residual intervention.

*New native MV-Adapter temporal comparison.*. We froze fifty Fresh C inputs not previously inspected through MV-Adapter, excluding all recorded earlier native and profile-panel objects. The input-only ordered selection used availability and native material/geometry compatibility. The native mesh normalization, six orthographic cameras and nvdiffrast rasterizer also generate evaluation targets; an independent source-UV sampling check validates this target path. Three textured new objects pass official/deployed and constant-default-path identity checks: 18 RGB images, conditioning features and injected residuals are bitwise identical. The old seven-object quality archive fails camera calibration and is superseded, not pooled.

The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; each stage-high schedule has 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five mean-matched controls each sum to 41.5 per scaled feature, within this interface. The fifty-object matrix is complete (350 rows), including thirteen texture-absent sources with white-albedo targets and a documented loader compatibility path. The other 37 textured objects are a descriptive subset, not a replacement primary cohort. Targets are unlit native-camera albedo, not Fresh C's differently normalized reference renders.

**Table 11. Native-camera MV-Adapter comparison, fifty input-selected objects. Views are averaged within object. Scale values and endpoints are specific to this interface, and are not numerical doses shared with GeoTex.**

| Condition | FG-PSNR | FG-LPIPS | FG-SSIM | Edge-SSIM |
|---|---|---|---|---|
| Native / Fixed High | 12.60 | 0.1280 | 0.5041 | 0.3526 |
| Fixed Low | 12.73 | 0.1270 | 0.5066 | 0.3475 |
| Fixed Mean | 12.68 | 0.1279 | 0.5068 | 0.3509 |
| Early-high | 12.60 | 0.1296 | 0.5041 | 0.3503 |
| Middle-high | 12.71 | 0.1262 | 0.5061 | 0.3470 |
| Late-high | 12.74 | 0.1260 | 0.5075 | 0.3514 |
| Generic Linear | 12.72 | 0.1263 | 0.5072 | 0.3501 |

The internally predeclared primary, Middle-high minus Fixed Mean FG-PSNR, is +0.029 dB (95% object-bootstrap CI [−0.019, +0.081]; paired Monte Carlo sign-flip $p$ = 0.281; 29/50 favorable). This does not support a stable middle-window gain or establish equivalence. The remaining 104 pair–metric tests use one Holm family. Middle-high has lower Edge-SSIM than Fixed Mean (adjusted $p$ = 0.00104); some other placement contrasts separate, including Late-high versus Early-high PSNR. Such endpoint-dependent differences do not demonstrate transferable C3 optimality. Complete contrasts, correction rules and the textured subset are in Supplementary Section S17. This is a native-interface validation of the schedul-

**Table 12. First-round 01549 3AFC results retained as reported. Percentages use 720 decisions per criterion; preference is not a measure of reference texture fidelity.**

| Method | Texture naturalness | Shape consistency | Overall quality |
|---|---|---|---|
| $s = 1.25$ | 332/720 (46.1%) | 96/720 (13.3%) | 176/720 (24.4%) |
| $s = 2.50$ | 104/720 (14.4%) | 331/720 (46.0%) | 126/720 (17.5%) |
| C3 (TCAS) | 284/720 (39.4%) | 293/720 (40.7%) | 418/720 (58.1%) |

ing question, not architecture-matched replication of GeoTex's capped residual intervention.

### *4.11. First-round cluster-aware preference study*

The 01549 first-round revision reported a blinded three-alternative forced-choice study on 30 objects with 24 participants. Participants compared conservative fixed-scale, aggressive fixed-scale, and C3 outputs for texture naturalness, shape consistency, and overall quality; condition order was randomized per trial. Each criterion had 720 first-choice decisions. Table 12 reports the aggregate counts; participant-level responses are not redistributed with this revision.

The reported analysis used a two-level participant/object bootstrap with 10,000 resamples. C3's overall-quality share was 58.1% (95% CI [50.1%, 65.8%]); its differences from the conservative and aggressive baselines were +33.6 points (CI [19.3, 47.2]) and +40.6 points (CI [29.4, 51.8]). The single-criterion advantages for texture naturalness and shape consistency had intervals crossing zero. This bounded preference result does not validate texture fidelity, unseen-view quality, or the Fresh C cohort.

### *4.12. FAC controlled re-examination reported in 01549*

A controlled re-examination of Full Adaptive Correction (FAC) was previously reported, with the base model, VAE, and GeoTex-Adapter frozen and roughly $6 \times 10^3$ controller parameters trainable. The stated protocol used the enhanced latent denoising objective (foreground weight 1.7, edge weight 1.3, and a latent-SSIM term), AdamW at peak learning rate $10^{-4}$ with 100 warm-up steps, and 2,000 training steps on a training pool reported as 1,706 objects and disjoint from evaluation. It reported a 300-object paired evaluation: every trained configuration reduced FG-PSNR versus TCAS by 1.10–2.94 dB, with at most 48/300 wins; the untrained C3-envelope control on 12 objects had $\Delta$FG-PSNR = $-0.18$ dB (95% CI $[-0.42, +0.05]$). The seven reported configurations and endpoint values are retained in Supplementary Table S16.

These FAC outcomes are retained as previously reported results. They are not presented as a new 1008 training run. The current archive audit found that a recovered configuration points to a 1,200-object list rather than the 1,706-object pool stated in 01549, and does not bind the training seed and run outputs in one manifest. We therefore disclose the remaining reproducibility gap and make no code-release claim; TCAS remains the sole training-free method.

## 5. Limitations

Fresh C is a held-out 300-object comparison whose conditions share one frozen checkpoint, runner, seed policy, inputs, metric implementation, and recorded native-cap semantics. The checkpoint is a controlled retraining and differs from the v1 path named for 01549 Table 4; the 01549 Table 9/FAC work uses a separate v2 strong-residual adapter. These historical v1/v2 differences are reported separately from the within-Fresh-C paired comparison. Fresh C does not numerically reproduce the old +0.96 dB Table 4 estimate. The historical maintable endpoint implementation is not recoverable for a numerical equivalence claim. Core-7 and the post-inspection strict-276 profile-placement analysis use the clean-v2 276-object list and the checkpoint SHA recorded for Fresh C, but are separate runs from Fresh C; Core-7 condition records do not establish one shared runner or complete input-tensor identity. They remain retrospective supporting diagnostics rather than confirmatory comparisons.

Texture variation and texture fidelity are distinct. The submitted RGB standard deviation, Laplacian variance, and gradient summaries are not re-used as fidelity measures; their source records and definitions are incomplete, and such metrics can rise with artifacts. Fresh C reports PSNR, LPIPS, and SSIM endpoints but has no validated texture-fidelity measure. The separate retrospective `FRESH_CONFIRM_300` analysis reports a 9/10 adjusted association family and quartile heterogeneity for LLH–GFL image-space deltas; it does not measure fidelity, establish causality, or predict which schedule should be used. The 01549 cluster-aware preference study is a separate subjective choice result, not a fidelity or 3D-quality endpoint. The earlier five-object bake has been corrected and remains separate from the ten-object compatibility-selected pilot. The ten-object Late-high–Fixed Mean LPIPS result is small and limited to unlit base-color evaluation; the C3–GFH PSNR advantage does not repeat stably. Input screening favors native-UV, simple-material objects, and cannot establish population-wide applicability. No validated seam or PBR measure is available. Historical N=20 unlit base-color renders provide only a small, mixed unseen-view result.

An exploratory four-object color-pathway audit found that changing the global appearance embedding can shift chroma. A separate Fresh B input-only audit found that stored embeddings matched neither of the two compared source-conditioning reconstructions within 0.03 for 150/150 inputs. This makes cache/source alignment a plausible contributor, not an established general cause of color drift. The C1 chroma-anchor candidate met its numeric gate (mean unseen-view $\Delta$FG-CIEDE2000 = $-1.5602$, 95% CI $[-2.2621, -0.6299]$, four of four) but failed the locked visual gate: one case showed no clear gain, and another retained mismatch with a faint cyan/blue rim. B2's selected raw-source embedding substitution did not meet its numeric gate (mean object-level change in the lower-is-better $\Delta$FG-CIEDE2000 endpoint = +0.1729, 95% CI $[-0.2044, +0.6810]$, one of four improved) and showed no clear visual correction. No independent Fresh B repair validation was performed. These development findings identify a control pathway and a plausible contributor, not a validated correction or a

complete root cause.

A separate, failure-enriched 26-object color diagnostic compared No Adapter, GFL, and LLH against paired six-view references. GFL reduced mean foreground CIEDE2000 by 19.65 (95% paired object-bootstrap CI $[-26.37, -13.25]$; 24/26 objects), while visible pink or purple shifts remained in two legacy cases whose original panel mapping is unavailable. LLH increased mean CIEDE2000 by 3.58 (95% CI $[+0.46, +6.71]$) while reducing local gradient and high-frequency variation. The selected set is not an independent confirmation cohort, and these measures do not establish texture fidelity. Figure 6 shows one of its 20 Fresh C panels, a different object from either legacy case. A decoded RGB snapshot showed one color shift before baking; a three-object P0 trace localized a cross-runtime divergence to the scheduler's scaling of the shared FP16 initial latent, but neither finding identifies the complete color-generation cause. No repair is validated (Supplementary Section S13).

The requested TCAS profile does not uniquely determine the applied scale or correction magnitude. Fresh C's native wrapper caps shallow requests at 0.8, and post-scale corrections depend on the adapter output. The strict-276 schedule sums are only approximately matched and do not equalize their layer-wise correction summaries. A3b's earlier raw-$L_2$ common-support comparison used feature groups with different dimensions, so it does not provide a dimension-normalized cross-layer dose-support test. These data do not identify a dose-independent mechanism, an optimal selector, or a schedule that dominates all endpoints and objects.

We withdraw the CAI optimality proposition and the earlier positive FAC extension claim; the negative FAC controlled re-examination and the cluster-aware 01549 preference study are retained as previously reported first-round results with their stated scope and provenance limits. The old pooled +0.96 dB headline and broad texture-retention claim are not used as current evidence. CLIP-IQA is not used to establish texture fidelity. The revised claim boundaries are documented in the changelog.

## 6. Conclusion

This study treats geometry-adapter strength as an observable inference control in a multi-view texturing pipeline. TCAS specifies a requested temporal profile on the existing GeoTex-Adapter path; C3, (1.25, 2.50, 1.25), is the recorded profile evaluated here, not a general scheduling rule.

On the matched 300-object Fresh C cohort, C3 exceeds GFL by +0.52 dB on the registered FG-PSNR endpoint (95% CI [+0.40, +0.64]), while GFH and Generic Linear have more favorable means than C3 on all four displayed endpoints. The applied deep/middle scale sums differ (82.5 for C3 and 62.5 for GFL; both have 40 in shallow), so this contrast does not isolate timing from applied scale. A separate post-inspection strict-276 placement analysis favors L–L–H over C3 across six endpoints; but does not match post-scale residual norms or provide an independent replication.

The equal-applied-budget 300-object matrix rejects a generally preferred middle window: C3 trails Fixed Mean and Late-high on all four image endpoints. In the calibrated ten-object bake pilot, Late-high improves unseen FG-LPIPS over Fixed Mean by 0.00265, favorable on all ten; this is a small named within-protocol result on compatible unlit assets. C3–GFH PSNR is not stably positive and SSIM is lower. Broad image/bake rank agreement remains positive, while the LPIPS winner changes on seven objects. The fifty-object native MV-Adapter primary does not separate Middle-high from Fixed Mean. These complete results support controlled empirical conclusions and interface/endpoint limits, not C3 optimality, a new scheduling primitive or general cross-backbone transfer. Seam quality, PBR fidelity, a validated selector and the full FAC training replay remain unresolved.

## Data availability

Object-level revision metrics, frozen protocols/configuration records, verified inference and analysis scripts, and the LaTeX source required to rebuild the manuscript and Supplement are included in the author-review package. The standalone reproduction archive supports statistic recomputation without model weights; fresh inference additionally requires the recorded weights, runtime and licensed source assets. Raw generated-image archives, trained checkpoints, source 3D assets whose redistribution rights are not established, and participant-level responses are not included. Additional analysis records may be requested from the corresponding author, subject to licensing, privacy, and archive constraints.

## Declaration of generative AI and AI-assisted technologies in manuscript preparation

OpenAI Codex assisted with revision drafting, implementation and reproducible statistical and figure code. OpenAI Images supplied conceptual artwork for Figure 1; the service did not expose a model version in the preserved generation record. Actual input/output thumbnails and experimental displays come from the recorded research runs and were not retouched with an image-generation tool. Numerical tables are computed from delivered object-level records. Final scientific review, accuracy and responsibility remain with the authors.

## Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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